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Worker Internship carried out at

DJEZZY
جازي

Students in the 3rd year of the Engineering Cycle
Telecommunications and Networking Systems

Authors:

ALAOUCHICHE Abderrahmane Yaakoub
KESSOUM Mohamed Walid

Internship Supervisor:

Malek GARET: Expert Radio and capacity

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Contents

1 Introduction **4**
1.1 Internship Objectives . . . 4
1.2 Company Overview . . . 4
1.2.1 Presentation . . . 4
1.3 Organizational Hierarchy . . . 5
1.3.1 Vision and Missions . . . 5

2 Radio Access Network **6**
2.1 Overview . . . 6
2.2 Workflow in RAN . . . 6
2.2.1 Department Roles and Collaboration . . . 7
2.2.2 Tools and Platforms . . . 7
2.2.3 Data Flow and Analysis . . . 8
2.3 Network Infrastructure . . . 8
2.3.1 Radio Access Network (RAN) . . . 8
2.3.2 Core Network (CN) . . . 10
2.3.3 Evolution of Mobile Generations . . . 10
2.3.4 Frequency Bands and Bandwidth Allocations . . . 11
2.4 Teams Overview . . . 11
2.5 Traffic & Budget Team . . . 11
2.6 Optimization Team . . . 12
2.6.1 Problem Detection and Resolution . . . 12
2.7 BSS Team . . . 12
2.7.1 Tracking and Location Management . . . 12
2.7.2 Cell Coding and Sectorization . . . 13
2.8 Radio Team . . . 14
2.8.1 Coverage Predictions and Optimization . . . 14
2.8.2 Difference Between Key LTE Radio Parameters . . . 15

3 Core Network **17**
3.1 Transmission Team . . . 17
3.1.1 Microwave Link Frequencies . . . 17
3.1.2 Modulation Techniques . . . 17
3.1.3 Microwave Link Setup . . . 18
3.1.4 Software: Pathloss . . . 19
3.1.5 Link Configuration and Supervision . . . 20
3.2 IP Team . . . 21
3.2.1 Overview of IP Core and IP RAN . . . 21
3.2.2 Network Architecture . . . 21
3.2.3 Internet Gateway (IGW) . . . 23
3.2.4 Network Architecture from IP Core to ISP . . . 23

4 Lab and MSC Visits **25**
4.1 Lab Visit 25
4.1.1 Evolution of Technology: From Dedicated Equipment to Software-Defined Radio 26
4.1.2 Key Components: Antenna, Remote Radio Unit (RRU), and Baseband Unit (BBU) 26
4.1.3 The Complete BSS Structure 29
4.1.4 Different Antennas from the Lab 30
4.2 MSC (Mobile Switching Center) Visit 32

5 Conclusion **33**

Chapter 1

Introduction

1.1 Internship Objectives

This internship was carried out at Djezzy, a major mobile network operator in Algeria specializing in telecommunications services, including mobile communication, network infrastructure, and customer solutions. This opportunity allowed us to gain a comprehensive understanding of the company's structure and the role of its various departments in ensuring seamless connectivity and service quality. A key focus was on deepening our knowledge of mobile communication infrastructure in Algeria, particularly how different departments collaborate to maintain and optimize network performance. Additionally, the internship provided insight into the evolution of telecommunications infrastructure across generations, allowing us to observe firsthand the technological advancements shaping the industry. Through this experience, we also gained practical exposure to network deployment, optimization strategies, and the integration of new technologies to enhance coverage and efficiency.

1.2 Company Overview

1.2.1 Presentation

The Algerian telecommunications operator Djezzy (officially known as Optimum Telecom Algerie, and formerly Orascom Telecom Algerie), founded in July 2001, is a leader in the mobile telephony sector in Algeria. With a subscriber base of over 15 million as of September 2023, Djezzy offers a full range of services, from prepaid to postpaid, including data, value-added services, and SUT. Djezzy has always stood out for its commitment to innovation and service quality, aiming today to position itself as a leading technology player in the country.

Since July 2022, a major shift has occurred with the transfer of nearly all VEON shares to the FNI (National Investment Fund). By becoming a Public Economic Enterprise, Djezzy is now 96.57% owned by the FNI, which plays a significant role in the country's socio-economic development. The company has set a goal to expand its commercial network nationwide, particularly in the Grand South, by opening stores in ten new provinces.

Djezzy has already achieved significant milestones in connectivity, covering 95% of the population across Algeria and successfully deploying 3G and 4G services in all 58 provinces. These key milestones in its history, such as obtaining 2G, 3G, and 4G licenses, illustrate its ongoing commitment to technological expansion.

Djezzy's ambition does not stop there. As a Public Economic Enterprise, it aims to become a major technological player in Algeria. Its objective is to promote the development of connectivity and actively contribute to the country's digitalization. With its experience and leadership, Djezzy strives to shape Algeria's technological future by offering innovative solutions and ensuring high-quality access to communication and connectivity nationwide.

1.3 Organizational Hierarchy

Djezzy's organizational structure is designed for seamless collaboration across specialized departments. This structure not only supports efficient operations in a highly competitive telecommunications market but also enables the company to swiftly adapt to technological advancements and customer needs.

Below is the global organizational hierachy of OTA :

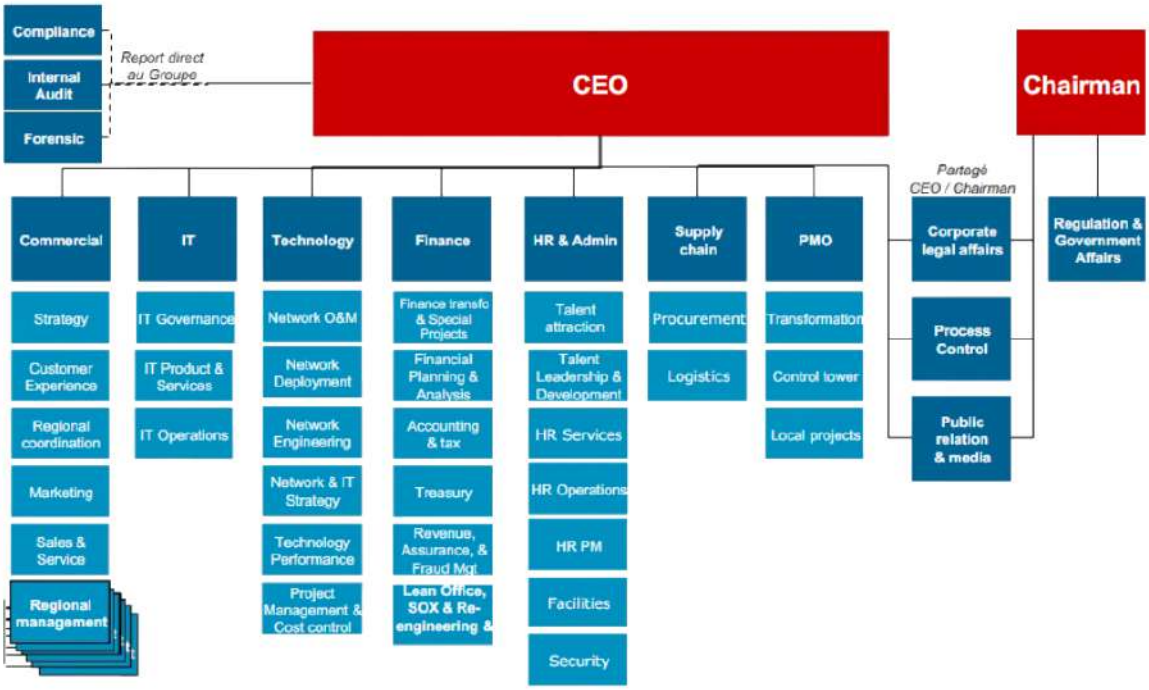

Figure 1.1: Organizational Hierarchy of the Company

1.3.1 Vision and Missions

Djezzy aims to be the preferred telecommunications operator in Algeria—a market leader that consistently delivers value to all its partners while setting the benchmark for workplace excellence and customer focus. The company's strategic priorities include:

- Delivering high-quality products and services at competitive prices.
- Deploying state-of-the-art technological infrastructures.
- Creating an exceptional work environment that promotes employee growth and satisfaction.
- Driving sustainable socio-economic development across Algeria.

Guided by its core values—commitment, teamwork, integrity, and innovation—Djezzy fosters a culture of excellence and responsibility in every aspect of its operations. In its mission to drive digital transformation and enhance connectivity, the company remains dedicated to continuous innovation and operational excellence. By leveraging its robust network and industry leadership, Djezzy strives to shape the future of Algeria's digital landscape.

Chapter 2

Radio Access Network

2.1 Overview

A Radio Access Network (RAN) is a crucial component of a mobile telecommunications system that connects user devices to the core network. It is responsible for handling communication between mobile devices and the network infrastructure, ensuring seamless data transmission, call management, and mobility. The RAN consists of multiple base stations that communicate wirelessly with mobile devices and manage radio resources efficiently.

In Djezzy, the RAN is also an essential department within the company. This department is responsible for overseeing the planning, deployment, and optimization of the radio network. Under the RAN department, there are various specialized subdivisions handling different aspects of network management and performance enhancement.

2.2 Workflow in RAN

As mentioned above, the Radio Access Network (RAN) department is composed of multiple teams working together to ensure the optimal performance, monitoring, and support of the network. These teams collaborate closely, each with specific responsibilities, as illustrated in Figure.

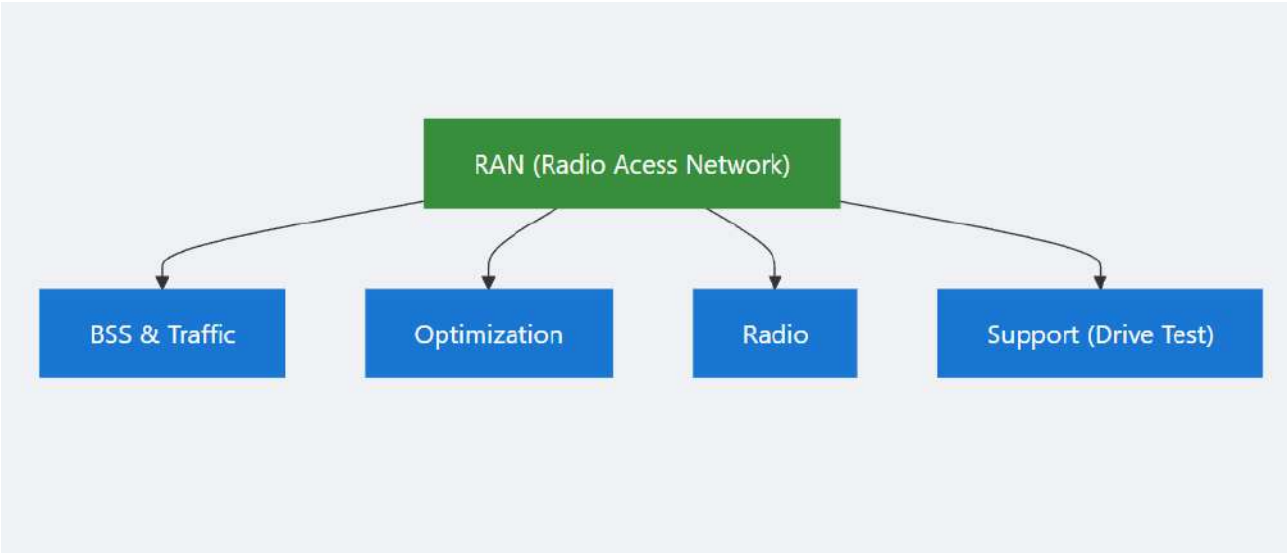

Figure 2.1: RAN Hierarchy

2.2.1 Department Roles and Collaboration

The RAN department comprises several specialized teams, each responsible for key aspects of network performance, capacity planning, and service quality. These teams work closely together to ensure a seamless and reliable user experience.

- **BSS:** The BSS team manages the transmission and reception of radio signals between mobile devices and the network. In addition to radio communications, they handle the configuration of new sites, network expansion planning, and resource allocation for each site. This process is performed in close collaboration with other teams to ensure optimal network coverage, capacity, and quality of service.
- **Traffic:** This team focuses on managing and optimizing network traffic capacity to meet growing subscriber demand for data and voice services. By monitoring and analyzing real-time traffic data, they ensure that the network can handle peak loads, especially during busy hours when usage is at its highest.
- **Optimization:** Responsible for enhancing network performance through KPI analysis, parameter tuning, and coverage improvement. Their objective is to continuously improve the quality of service by minimizing network issues and maximizing user experience.
- **Radio:** This team ensures reliable wireless coverage and service quality. Engineers are assigned to specific regions and are tasked with monitoring, maintaining, and improving coverage, especially in areas with weak or no signal.
- **Support (Drive Test):** Conducts field measurements and drive tests to assess network performance, validate coverage, and identify areas for improvement. Their data plays a vital role in optimizing and troubleshooting network issues.

These teams collaborate through shared data, joint analyses, and coordinated efforts to ensure the continuous improvement and expansion of the network infrastructure.

2.2.2 Tools and Platforms

The RAN department utilizes a variety of tools provided by multiple vendors to monitor, analyze, and manage network performance. While these tools are developed by different vendors, they are not limited to specific functionalities and can be used across multiple tasks such as monitoring, optimization, and data tracing. Table 2.1 summarizes the main tools used in the department.

Vendor	Tools
Nokia	NetAct, NPM, MegaMon, DCAP
ZTE	UMEI, NetMax
Huawei	U2020, PRS (MAOS), Discovery

Table 2.1: Tools Used in the RAN Department

These platforms are integrated with Djezzy's servers, which continuously transmit raw counters and key performance indicator (KPI) data. Engineers can extract and analyze this data across various dimensions such as time (daily, weekly, hourly), region, and network generation (2G, 3G, 4G). The tools also support the creation of customized reports and offer visualization capabilities to monitor trends and detect deviations from acceptable performance standards. Below are some screenshots of some of the tools :

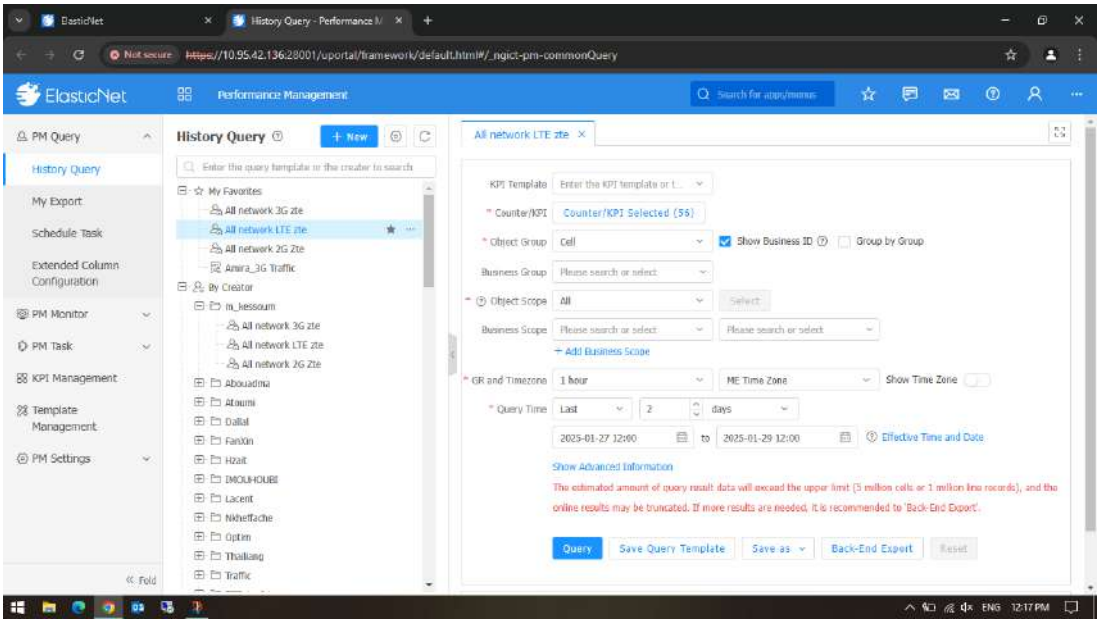

Figure 2.2: UMEI Tool

2.2.3 Data Flow and Analysis

These platforms are connected to Djezzy's multiple servers, which continuously transmit raw counters and key performance indicator (KPI) measurements. Engineers can extract and analyze these KPIs based on various parameters such as time (daily, weekly, hourly), region, and network generation (2G, 3G, 4G).

Data Extraction and Reporting

The tools allow engineers to generate personalized reports tailored to specific needs, including the detection of anomalies and performance deviations. These reports are crucial for identifying and addressing network problems efficiently.

Visualization and Trend Monitoring

Built-in visualization tools enable engineers to monitor trends, detect deviations from acceptable performance limits, and take corrective actions. This helps ensure continuous improvement of network performance and reliability.

2.3 Network Infrastructure

The company's network infrastructure supports multiple generations of mobile communication technologies, including 2G (GSM), 3G (UMTS), and 4G (LTE). Each generation enhances network performance, offering improved services, higher data speeds, and better efficiency.

2.3.1 Radio Access Network (RAN)

The RAN is responsible for wireless communication between user devices and the core network. It consists of different components for various technologies:

2G (GSM)

- **BTS (Base Transceiver Station)**: Manages radio communication with mobile phones.

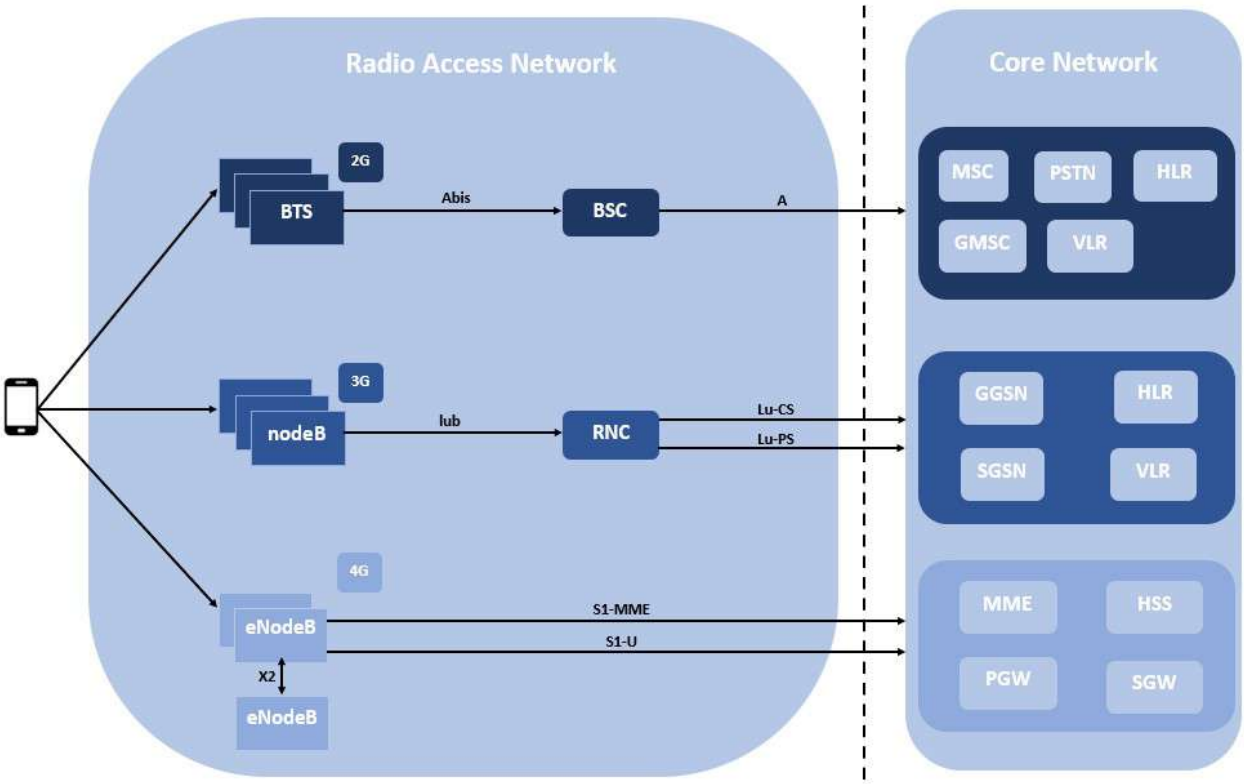

Figure 2.3: Overview of Network Infrastructure Across Generations

- **BSC (Base Station Controller)**: Controls multiple BTS units, managing radio resources and handovers.
- **Interfaces**:
  - **Abis**: Connects BTS to BSC.
  - **A**: Connects BSC to the Core Network.

3G (UMTS)

- **NodeB**: Equivalent to BTS but for 3G networks.
- **RNC (Radio Network Controller)**: Manages multiple NodeB units and controls mobility.
- **Interfaces**:
  - **Iub**: Connects NodeB to RNC.
  - **Lu-CS**: Connects RNC to the Core Network for voice.
  - **Lu-PS**: Connects RNC to the Core Network for data.

4G (LTE)

- **eNodeB**: The base station in LTE, integrating NodeB and RNC functions.
- **Interfaces**:
  - **X2**: Communicates between eNodeBs.
  - **S1-MME**: Connects eNodeB to MME for signaling.
  - **S1-U**: Connects eNodeB to SGW for data transmission.

2.3.2 Core Network (CN)

The Core Network handles call switching, data routing, authentication, and mobility management. It consists of:

Circuit-Switched Core (2G/3G Voice)

- **MSC (Mobile Switching Center)**: Manages call setup and routing.
- **PSTN (Public Switched Telephone Network)**: Connects to landlines.
- **HLR (Home Location Register)**: Stores permanent subscriber information.
- **VLR (Visitor Location Register)**: Stores temporary subscriber data for roaming.

Packet-Switched Core (3G/4G Data)

- **GGSN (Gateway GPRS Support Node)**: Connects the mobile network to the Internet.
- **SGSN (Serving GPRS Support Node)**: Manages mobile sessions and data flow.
- **HSS (Home Subscriber Server)**: Stores subscriber profiles for 4G.
- **MME (Mobility Management Entity)**: Manages authentication and handovers.
- **PGW (Packet Data Network Gateway)**: Routes user traffic to external networks.
- **SGW (Serving Gateway)**: Routes user data between RAN and PGW.

2.3.3 Evolution of Mobile Generations

Each new mobile generation introduced significant technological advancements, impacting connectivity, data speeds, and global communication. Below are key differences between generations:

- **2G (GSM)**: Enabled digital voice calls and SMS, replacing analog systems and paving the way for global mobile communication.
- **3G (UMTS)**: Introduced mobile internet access, supporting video calls, web browsing, and multimedia messaging.
- **4G (LTE)**: Delivered high-speed mobile broadband, enabling seamless streaming, real-time applications, and the foundation for modern digital services.

The transition between these generations revolutionized the way people communicate, consume content, and conduct business, laying the groundwork for future innovations such as 5G and beyond.

2.3.4 Frequency Bands and Bandwidth Allocations

The evolution of mobile communication has been driven by advancements in frequency utilization and bandwidth efficiency. The table below summarizes the frequency bands allocated across different mobile generations and that is specifically used by OTA Operator:

Table 2.2: Frequency Bands Across Mobile Generations

Generation	Frequency Bands
4G LTE	900 MHz (FDD), 1800 MHz (FDD), 2100 MHz (FDD), 2300 MHz (TDD)
3G UMTS	900 MHz, 1800 MHz, 2100 MHz
2G GSM	900 MHz, 1800 MHz

The corresponding bandwidth allocations for each frequency band are detailed below:

Table 2.3: Bandwidth Allocations Across Frequency Bands

Frequency Band (MHz)	Allocated Bandwidth (MHz)
900	8
1800	18
2100	20
2300	30
Total Spectrum	**76**

2.4 Teams Overview

2.5 Traffic & Budget Team

This team plays a pivotal role in managing and optimizing the traffic capacity within the overall network infrastructure. It is responsible for ensuring that the access network effectively supports the ever-growing data and voice services demanded by subscribers. By monitoring and analyzing real-time and historical traffic data, the department ensures that the network can efficiently handle the peak loads, particularly during the busy hour of the day—the period when network usage reaches its highest intensity.

The busy hour is a critical metric for capacity planning, as it directly influences both the capacity and budget allocation for circuit-switched (CS) and packet-switched (PS) data services. During this peak period, the highest traffic levels are observed, which informs the necessary adjustments and expansions required to maintain optimal performance and service quality.

Capacity studies in the department are methodically conducted by analyzing average (moyenne) traffic levels combined with Key Performance Indicators (KPIs) across various periods. These studies are particularly focused on different temporal scenarios such as: Normal Periods: Regular traffic patterns that establish baseline performance metrics. Ramadan: A period with unique behavioral patterns where traffic usage may vary significantly due to cultural and social factors. Summer: Seasonal variations that can lead to increased network usage due to holidays and travel.

Through these detailed capacity studies, the Access Network department can proactively adjust network resources, ensuring that both CS and PS infrastructures are sufficiently equipped to handle current and projected demands. This strategic planning not only secures uninterrupted service delivery but also optimizes budget allocations, ensuring investments in network expansion and technological upgrades are both efficient and effective.

2.6 Optimization Team

The Optimization department plays a crucial role in improving network performance and ensuring that key performance indicators (KPIs) meet the required standards. This department is organized by wilayas, with each optimization engineer assigned to specific regions. Each engineer is responsible for optimizing the KPIs of one or two vendors from Djezzy's three vendors: Huawei, Nokia, and ZTE. Collaboration between engineers is essential to ensure cohesive optimization across the network.

As seen in the figure above, this is the UME platform from ZTE. The platform offers a variety of options, including Performance Management, RAN Configuration, and Signaling Trace, among others. In the Performance Management section, we can view all public reports generated by engineers using this platform. These reports are accessible for review, and the data can be exported for further analysis. Additionally, the platform allows users to create custom reports tailored to specific needs, providing flexibility for engineers to address particular performance issues.

2.6.1 Problem Detection and Resolution

Problems can be identified through customer complaints received from customer service, a significant drop in general KPIs, or issues arising from new installations or modifications in the network. Once a problem is detected, engineers follow a structured process: they analyze the issue using the relevant platform to identify its root cause, generate and test commands within the platform to resolve it, create a work order for the execution team to review and implement the commands, and finally, recheck the KPIs post-implementation to ensure the problem is resolved.

Example: Resolving Overshooting Issues

During our visit, an engineer shared a case where a site was overshooting, meaning its signal extended beyond the intended coverage area and failed to properly interact with its neighboring cells. This led to poor KPIs, such as a high "Attempt Fail Rate" rate. The solution involved adjusting the mechanical tilt of the site's antennas to reduce overshooting and improve its interaction with neighboring cells. After implementing the adjustment, the KPIs were re-evaluated, showing significant improvement in performance.

2.7 BSS Team

The BSS team is responsible for configuring new sites, planning network expansions, and assigning the appropriate resources to each site, in addition to handling radio communications resources. This process is performed in close collaboration with other teams to ensure optimal network coverage, capacity, and quality of service. The integration between these teams ensures that new and existing sites are meticulously designed to meet current and future network demands.

During discussions with workers in the BSS Team, we gained the following insights into their operations and processes.

2.7.1 Tracking and Location Management

Efficient tracking and location management within a cellular network are achieved using various codes and lists, including the Tracking Area Code (TAC), Location Area Code (LAC), and Tracking Area List (TAL).

- **Tracking Area Code (TAC)**

  - Used in 3G and 4G networks to identify a specific tracking area where a user is located.
  - Enables the network to manage user mobility and optimize resource allocation.
  - **Example**: A TAC can correspond to a city or a broader region, allowing the network to monitor user locations within that area efficiently.

- **Location Area Code (LAC)**
  - Primarily utilized in 2G and 3G networks to define a geographical area for location management.
  - Helps the network efficiently manage paging and handovers.
  - **Example**: An LAC may be assigned to a district or a cluster of cities.

- **Tracking Area List (TAL)**
  - Applied in LTE (4G) networks, TAL comprises multiple TACs to optimize network tracking.
  - Reduces frequent location updates, thereby minimizing signaling load and enhancing efficiency.
  - **Example**: A TAL can include several TACs covering multiple adjacent cities or regions, enabling efficient user tracking across broader areas.

2.7.2 Cell Coding and Sectorization

Cell Coding

The cell coding format follows this structure:

- `4A16X001_1`, `3O20M093_2`, `C25X128_3`

Structure Breakdown:

- **4/3**: Generation (4th/3rd generation)
- **A/O/C**: Geographical division (Radio Center, West, East)
- **Wilaya Code**: From 01 to 48 (mapping for 10 extra in progress)
- **X/M**: Cell type (Macro/Micro)
- **Site Number**: Specific site identifier
- **Cell Number**: The underscore number specifies the cell name.

Sectorization

Cells are sectorized based on frequency, following a clockwise pattern on a site to manage capacity and coverage.

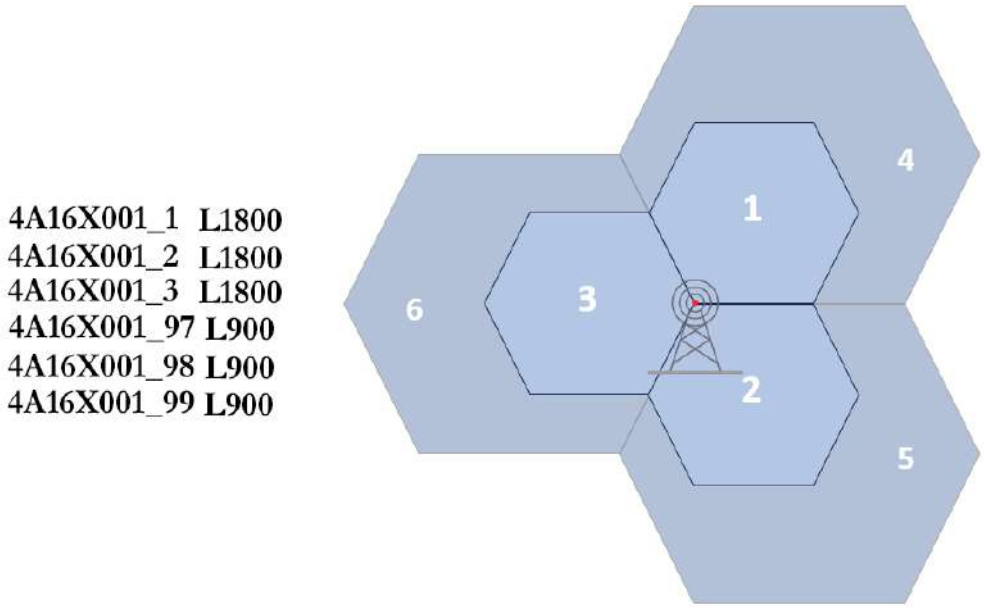

Figure 2.4: Example of Sectorization for Cells on a Site

In Figure 2.4, a site is divided into six sectors, each assigned a frequency band. The listed sectors (1, 2, 3) operate on L1800, while (97, 98, 99) use L900. Higher frequencies (L1800) provide greater capacity, while lower frequencies (L900) ensure better coverage.

This sectorization approach enhances network efficiency by optimizing spectrum usage, reducing interference, and increasing user capacity.

2.8 Radio Team

The Radio department is responsible for managing and optimizing the wireless network to ensure reliable coverage and service quality. Engineers are assigned to different wilayas, each responsible for maintaining and improving coverage in their respective areas. Their primary task is to ensure good signal availability, particularly in locations with weak or no coverage.

The process begins with planning new site deployments based on customer complaints, coverage gaps, and capacity demands. Engineers identify the best locations for new sites, considering factors such as terrain, interference, and population density. They also conduct field visits to oversee site implementation and ensure that the network is optimized once the site becomes operational.

To assist in this planning, engineers use radio network simulation tools like **Atoll**. This software allows them to visualize signal propagation, predict coverage quality, and analyze potential interference before deploying new sites.

2.8.1 Coverage Predictions and Optimization

The coverage prediction shown in Figure 2.5 illustrates the expected signal strength for a newly planned site. The different colors represent various signal levels, with green and dark green indicating strong coverage, while red and pink highlight areas with weaker or no signal. This visualization helps engineers assess whether a site provides adequate coverage and whether adjustments are needed.

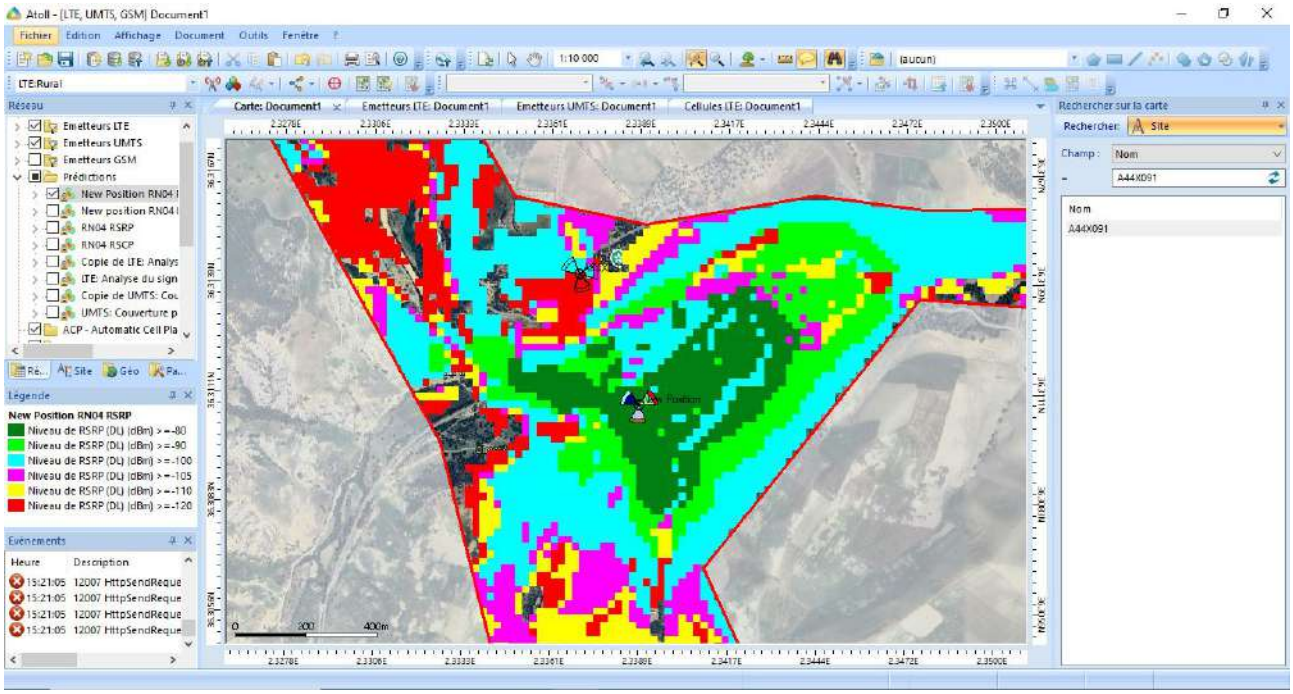

Figure 2.5: Coverage Prediction in Atoll for a New Site Deployment

By analyzing these predictions, engineers can optimize antenna positioning, adjust power levels, and fine-tune configurations to improve network performance. This approach ensures efficient site deployment, reduces interference, and enhances user experience by providing better signal quality and connectivity.

2.8.2 Difference Between Key LTE Radio Parameters

Reference Signal Received Power (RSRP)

RSRP is the power level of the LTE reference signals received by the user equipment (UE). It helps in determining the coverage area of an LTE cell.

$$\text{RSRP} = \frac{\sum P_{RS}}{N} \tag{2.1}$$

where P_{RS} is the power of the reference signals, and N is the number of resource elements used for reference signals.

Reference Signal Received Quality (RSRQ)

RSRQ measures the quality of the received reference signals. It considers both RSRP and the total received power (RSSI), providing an indicator of interference and congestion.

$$\text{RSRQ} = \frac{N \times \text{RSRP}}{\text{RSSI}} \tag{2.2}$$

where N is the number of resource blocks used.

Received Signal Strength Indicator (RSSI)

RSSI is the total received power, including all interference and thermal noise. It gives an overall measure of signal strength.

$$\text{RSSI} = P_{\text{total}} = P_{\text{RSRP}} + P_{\text{interference}} + P_{\text{noise}} \tag{2.3}$$

Signal-to-Noise-and-Interference Ratio (SNIR)

SNIR represents the ratio of useful signal power to the combined power of noise and interference. It is crucial for estimating data throughput and link quality.

$$\text{SNIR} = \frac{P_{\text{signal}}}{P_{\text{interference}} + P_{\text{noise}}} \tag{2.4}$$

Higher SNIR values indicate better signal quality and improved network performance.

Chapter 3

Core Network

3.1 Transmission Team

The Transmission department is crucial for ensuring reliable communication over long distances. It manages the transport of data through various frequencies and technologies, enabling seamless connectivity across the network.

3.1.1 Microwave Link Frequencies

The department uses a range of frequencies for microwave links, each with specific characteristics in terms of distance and antenna dimensions:

Frequency Band	**Minimum Distance (km)**	**Maximum Distance (km)**	**Antenna Size (m)**
15 GHz	5	40	1.8
18 GHz	4	15	1.2
26 GHz	2	7	0.9
38 GHz	0.5	3.5	0.6
42 GHz	0.3	2	0.3
E-band (80 GHz)	Primarily used for high-capacity short links		

Table 3.1: Frequency Bands, Distances, and Antenna Sizes for Microwave Links

The choice of frequency depends on the distance, required throughput, and environmental factors. Lower frequencies (e.g., 15 GHz) are used for longer distances, while higher frequencies (e.g., 42 GHz and E-band) are suited for short, high-capacity links.

3.1.2 Modulation Techniques

Microwave links use adaptive Quadrature Amplitude Modulation (QAM) to optimize performance based on channel conditions. Supported modulation schemes include:

- **4QAM, 16QAM, 64QAM, 128QAM, 1024QAM, and 4096QAM.**

Higher QAM levels (e.g., 1024QAM, 4096QAM) provide higher data rates but are more susceptible to errors due to noise and interference. Adaptive modulation ensures the system adjusts dynamically to maintain a balance between performance and reliability.

	7Mhz	14Mhz	28Mhz	56Mhz	112Mhz	250Mhz	500Mhz	1000Mhz
QPSK	11Mbit/s	37Mbit/s	43Mbit/s	89Mbts	180Mbts	368Mbit/s	740Mbit/s	1553Mbts
16QAM	21Mbit/s	42Mbit/s	86Mbit/s	180Mbts	361Mbts	759Mbit/s	1527Mbit/s	3173Mbts
32QAM	26Mbit/s	53Mbit/s	110Mbit/s	217Mbit/s	452Mbts	949Mbit/s	1850Mbit/s	3886Mbts
64QAM	33Mbit/s	66Mbit/s	136Mbit/s	273Mbit/s	575Mbts	1107Mbit/s	2224Mbit/s	3594Mbit/s
128QAM	38Mbit/s	78Mbit/s	161Mbit/s	323Mbit/s	677Mbts	1329Mbit/s	2672Mbit/s	5440Mbts
256QAM	43Mbit/s	89Mbit/s	183Mbit/s	368Mbit/s	780Mbts	1519Mbit/s	3054Mbit/s	6217Mbts
512QAM	49Mbit/s	94Mbit/s	196Mbit/s	395Mbit/s	891Mbts	1709Mbit/s	3436Mbit/s	
1024QAM	54Mbit/s	104Mbit/s	217Mbit/s	447Mbit/s	956Mbts			
2048QAM		129Mbts	242Mbit/s	501Mbit/s	1049Mbts			
4096QAM			290Mbts	583Mbts	1123Mbts			

Figure 3.1: Modulation Frequency

This table shows the relationship between modulation schemes, channel bandwidth, and data rates. Modulation ranges from QPSK to 4096QAM, where higher schemes provide significantly higher data rates but require better channel conditions (e.g., higher SNR). Channel bandwidths range from 7 MHz to 1000 MHz, with wider bandwidths enabling greater throughput. For instance, at 7 MHz, QPSK achieves 11 Mbit/s, while 4096QAM reaches 290 Mbit/s; at 112 MHz, QPSK delivers 180 Mbit/s compared to 1123 Mbit/s with 4096QAM. This highlights the importance of adaptive modulation, which dynamically balances throughput and reliability based on channel quality.

3.1.3 Microwave Link Setup

The setup of a microwave link involves the following components:

- **Indoor Unit (IDU)**: Processes the data and converts it to an intermediate frequency (IF).
- **Cable**: Transmits the IF signal to the outdoor unit.
- **Outdoor Unit (ODU)**: Converts the IF signal to the microwave frequency.
- **Antenna**: Transmits the signal to the destination.
- **Channel and Remote Site**: A similar setup exists on the other side, ensuring bidirectional communication.

The installation must respect the **line of sight (LOS)** and **Fresnel zone clearance** to avoid signal obstruction.

Figure 3.2: Microwave Link Setup Example

3.1.4 Software: Pathloss

The *Pathloss* software is used to design and configure microwave links. It allows engineers to simulate link performance, calculate link budgets, and ensure optimal alignment and configuration.

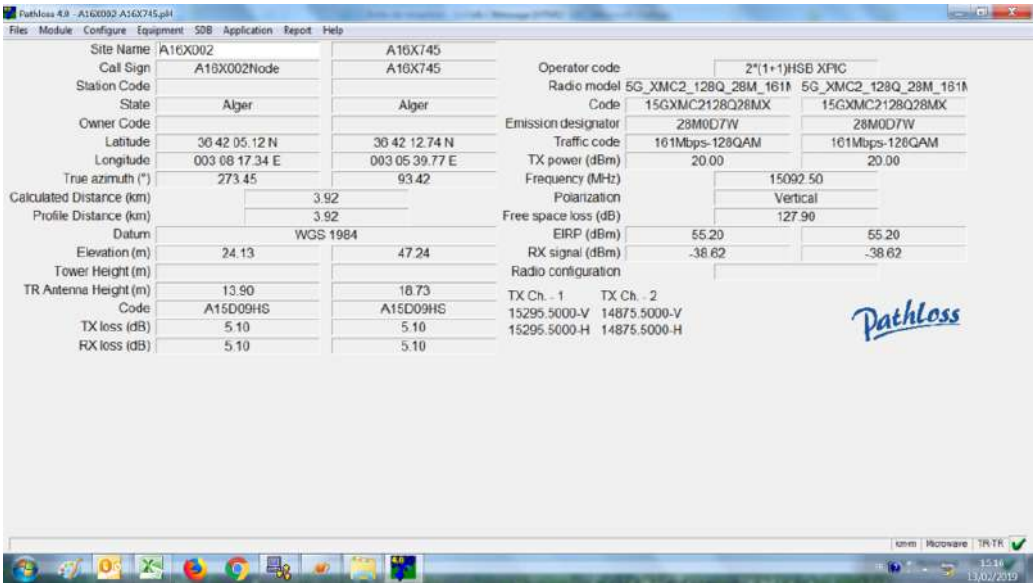

Figure 3.3: Pathloss Software Interface

This Pathloss configuration connects two sites, A16X002 and A16X745, over a 3.92 km link using the 15 GHz band. The link employs 128QAM modulation, delivering 161 Mbps per channel. Key parameters include a TX power of 20 dBm, RX signal of -38.62 dBm, free space loss of 127.90 dB, and EIRP of 55.20 dBm. Antennas are installed at heights of 13.90 m and 18.73 m. The link is well-optimized for its distance, with strong signal strength and reliable performance, suitable for high-capacity communication.

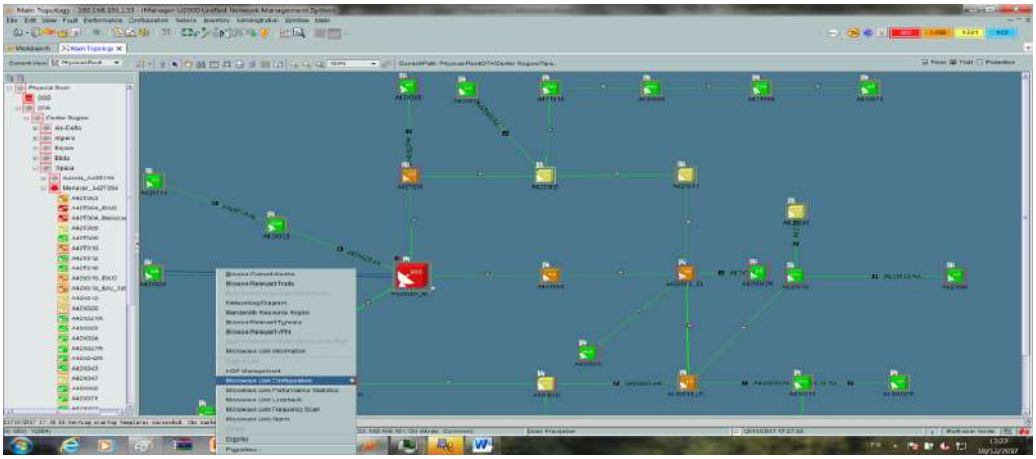

Figure 3.4: Microwave link supervision A42T004-A42X024

3.1.5 Link Configuration and Supervision

Figure 3.4 shows the microwave link **A42T004-A42X024** in the network management system. The interface provides real-time monitoring of the link's status, with color-coded icons indicating its condition, helping to quickly identify issues. This supervision ensures reliable transmission and efficient resource management.

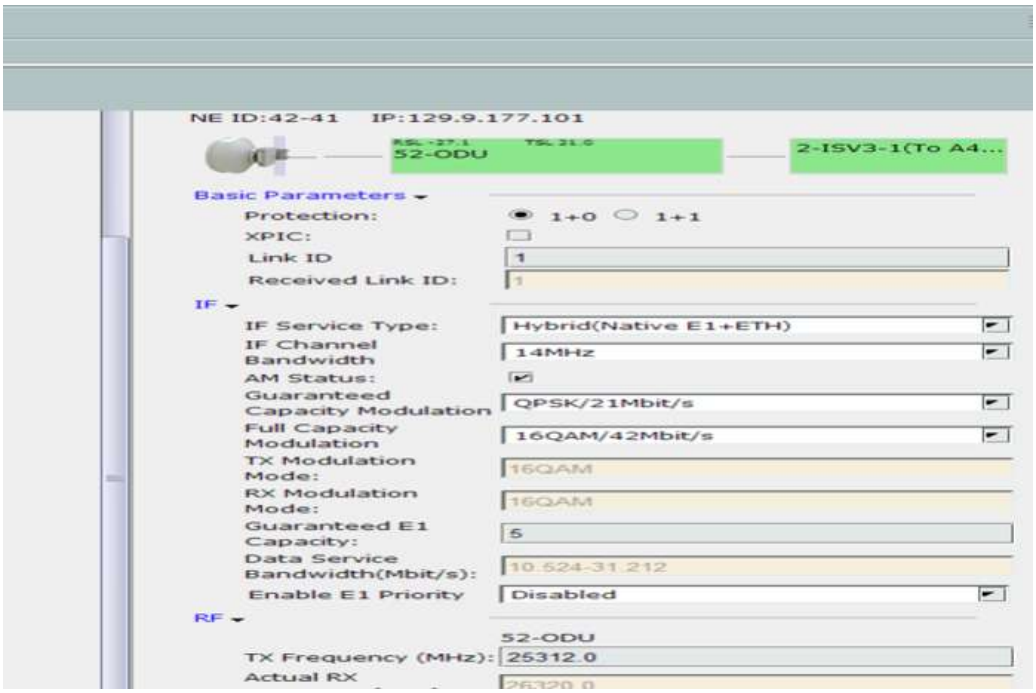

Figure 3.5: Microwave link configuration A42T004-A42X024

Figure 3.5 displays the configuration settings for the link **A42T004-A42X024**, where parameters like bandwidth, modulation type, and transmission capacity can be adjusted. Configured in **hybrid mode (Native E1 + Ethernet)**, it supports both legacy and modern IP transmission. Proper configuration is key to balancing performance and link reliability.
Transmission link design balances capacity needs with minimal equipment usage.

3.2 IP Team

3.2.1 Overview of IP Core and IP RAN

The IP RAN (Radio Access Network) is directly connected to the IP Core, which consists of interconnected routers. The Internet Gateway (IGW) is also connected to the IP Core, and the IT infrastructure links to the IGW. Ultimately, all of these systems are connected to the ISP, Algérie Télécom, forming a comprehensive network.

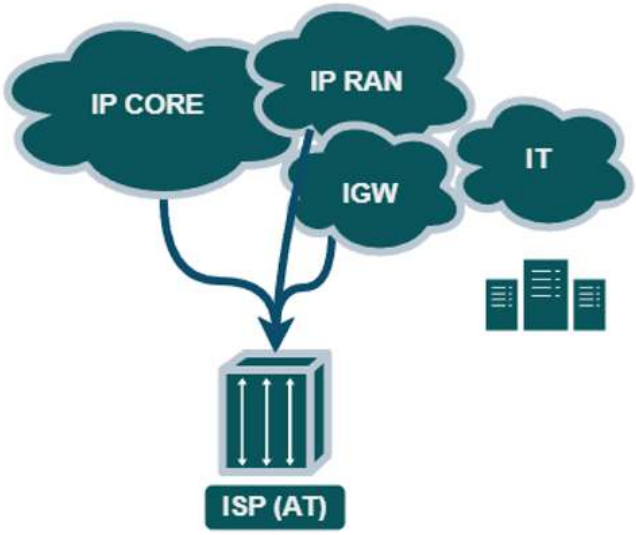

Figure 3.6: Globale IP Network

3.2.2 Network Architecture

The IP Core is designed with a highly redundant and interconnected backbone network. Djezzy utilizes nearly 11 rings within the IP Core, ensuring continuous service availability. These rings are connected to the backbone routers, which play a crucial role in maintaining the network's robustness and reliability.

Figure 3.7: Current Ring Architecture

Core Routers

Within the IP Core architecture, there are Provider Routers (P routers) and Provider Edge Routers (PE routers). The P routers form the backbone of the network, interconnected to ensure high data throughput and resilience. The PE routers, located at the edges of the network, are configured in pairs for redundancy, with two routers always present for failover scenarios.

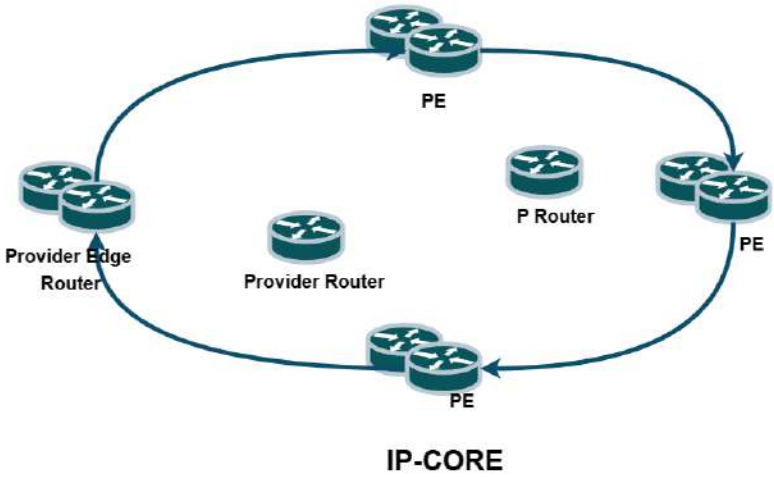

Figure 3.8: IP Core Routers Architecture

Redundancy and Routing The interconnection of routers is designed with multiple pathways for data transmission, ensuring that if one route fails, another is available. This dual-path setup is known as the "Plan A and Plan B" approach. The use of redundant paths minimizes the risk of network downtime and enhances overall stability.

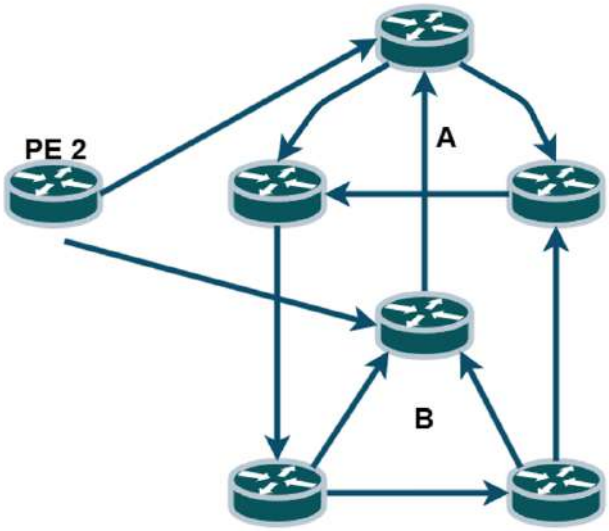

Figure 3.9: Redundancy Plans

Routing Protocols

In the Provider Edge (PE) routers, various routing protocols are employed to ensure efficient data routing and network management. These protocols include:

- **Static Routing**
- **OSPF** (Open Shortest Path First)

- **ISIS** (Intermediate System to Intermediate System)
- **RIP** (Routing Information Protocol)
- **BGP** (Border Gateway Protocol)

Additionally, each PE router is aware of all other PE routers within the network, facilitating dynamic routing and fault tolerance.

3.2.3 Internet Gateway (IGW)

The Internet Gateway (IGW) is a critical component that connects the IP Core to the broader internet. Before traffic from the IP Core reaches the IGW, it passes through a set of firewalls. These firewalls ensure the security of the network by filtering inbound and outbound traffic. Like other components, two firewalls are deployed for redundancy, ensuring that if one fails, the other can continue to protect the network.

At the termination point of the IGW, there are three main exit points connecting to the Algérie Télécom ISP: Annaba, Oran, and Algiers. These exit points serve as the primary routes for Djezzy's internet traffic to reach the external world.

NAT Protocol at the IGW At the IGW level, the NAT (Network Address Translation) protocol is used to translate internal private IP addresses to public ones, enabling multiple devices within the Djezzy network to share a smaller set of public IP addresses. This optimizes IP address allocation and ensures secure communication with external networks.

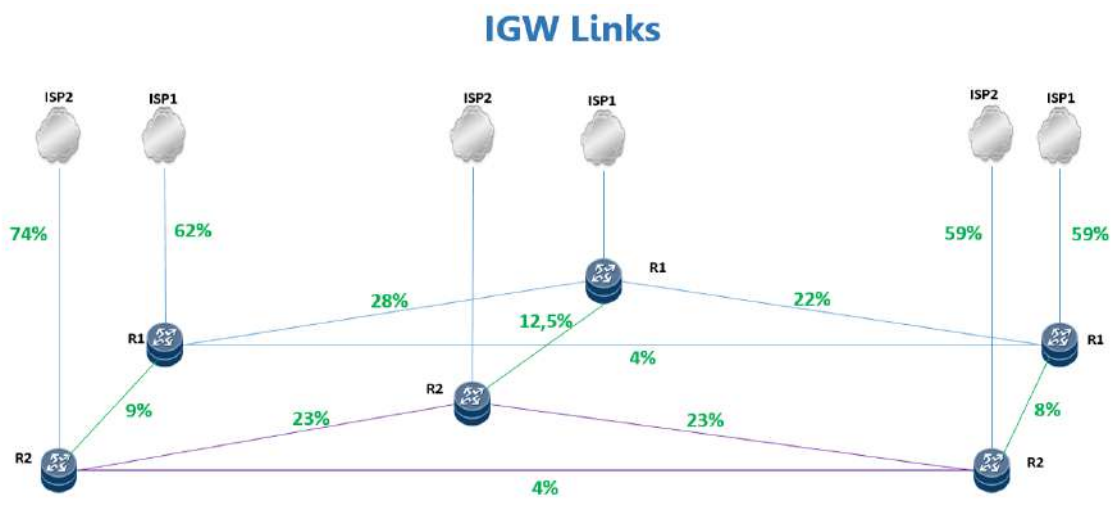

Figure 3.10: IGW Links and Connectivity

3.2.4 Network Architecture from IP Core to ISP

To better understand the flow of data from the IP Core to the ISP, the diagram below provides a clear representation of the network architecture, illustrating how the IP Core connects to the ISP through various components.

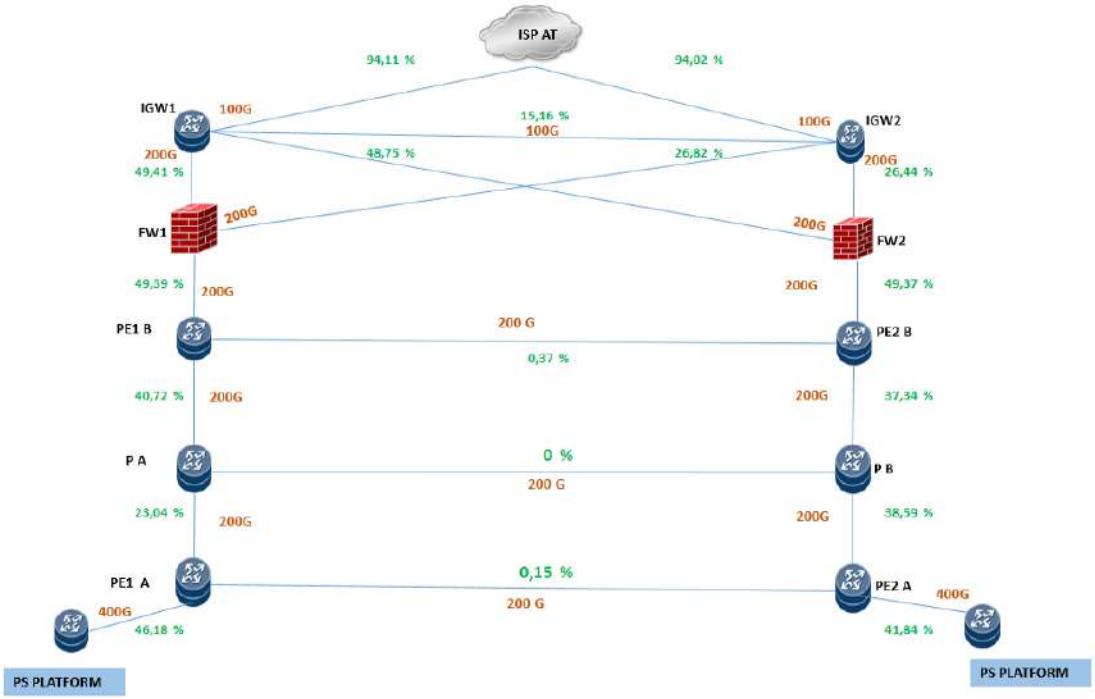

Figure 3.11: Architecture from the IP Core to the ISP

In summary, Djezzy's IP Core architecture is built with redundancy and high availability in mind. The network is designed to ensure efficient data transmission and fault tolerance through multiple layers of backup systems, including routers, firewalls, and exit points to the ISP. This robust architecture ensures that Djezzy can maintain reliable service for its users.

Chapter 4

Lab and MSC Visits

4.1 Lab Visit

The laboratory is a critical space for testing and innovation in telecommunications. During our visit, we had the opportunity to observe the dynamic environment where new technologies and solutions are rigorously tested. The focus of this lab is on experimentation, enabling engineers to explore and evaluate various advancements in the field.

Figure 4.1: Laboratory

During our visit, we were accompanied by a BSS Planning Engineer who provided an in-depth overview of the lab's primary focus: testing and experimentation. The lab is dedicated to conducting tests and trialing new technologies in modern telecommunications systems. Engineers constantly experiment with various solutions to assess their performance and impact. Below, we detail the insights gained from this hands-on approach to innovation.

4.1.1 Evolution of Technology: From Dedicated Equipment to Software-Defined Radio

Initially, each generation of wireless technology (2G, 3G, 4G) required dedicated equipment. This posed challenges in terms of scalability and cost-efficiency. However, with the advent of **Software-Defined Radio (SDR)**, these issues have been resolved. SDR allows a single piece of equipment to support multiple technologies simultaneously, enabling a more streamlined and cost-effective network architecture.

4.1.2 Key Components: Antenna, Remote Radio Unit (RRU), and Baseband Unit (BBU)

Modern telecommunications infrastructure consists of three main hardware components: the **antenna**, the **Remote Radio Unit (RRU)**, and the **Baseband Unit (BBU)**. Each plays a distinct and critical role in the network.

The Antenna

The antenna plays a vital role in transmitting and receiving electromagnetic signals, supporting various configurations depending on the technology and coverage requirements. Common setups include:

- **2T2R**: Two transmitters and two receivers.
- **4T4R**: Four transmitters and four receivers.
- **64T64R**: Advanced configurations used in modern Massive MIMO systems for 5G.

Antenna polarization is equally important, as it determines how electromagnetic waves are transmitted and received. Most modern cellular antennas utilize **dual slant polarization** or **cross-polarization**, where signals are transmitted at orthogonal angles (e.g., +45° and -45°). This technique reduces interference, enhances signal quality, and supports multiple-input multiple-output (MIMO) systems effectively.

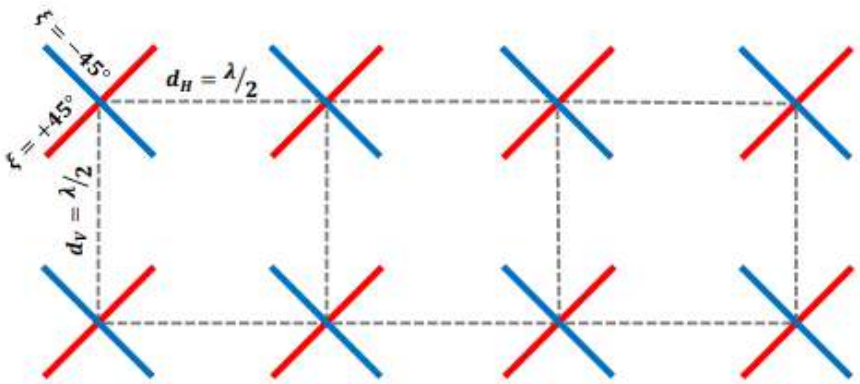

Figure 4.2: Uniform Planar Arrays (UPA) with cross-polarized antenna elements

Figure 4.3: ZTE AAU

Remote Radio Unit (RRU)

The RRU, also known as the Radio Frequency Unit, is the intelligent component responsible for:

- Performing modulation and demodulation of signals.
- Converting digital signals to analog (and vice versa).
- Amplifying the signals for transmission.

The placement of the RRU is crucial to minimize signal losses:

- When the RRU is placed at the base of the tower and connected to the antenna via long guided cables, signal losses increase significantly.

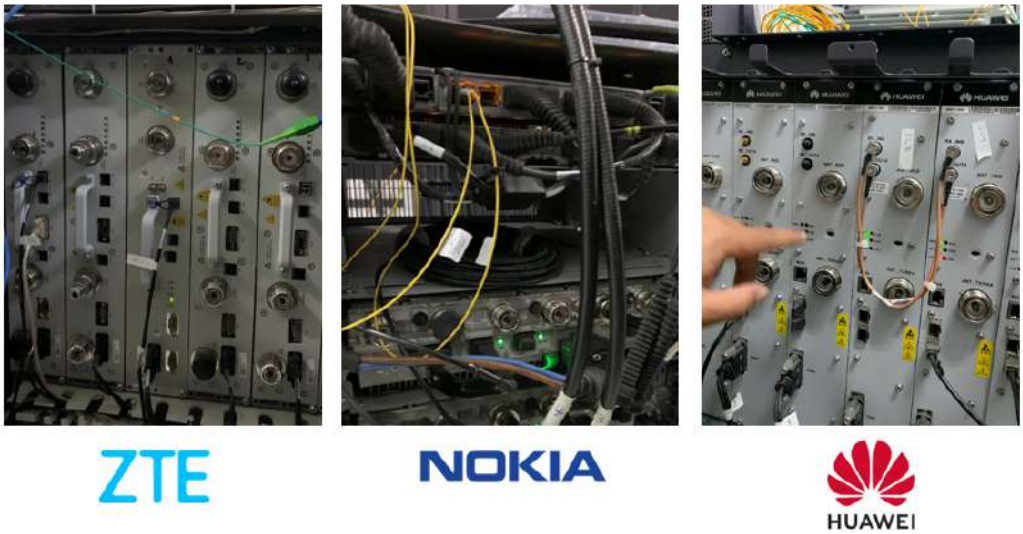

Figure 4.4: RFU

- Placing the RRU close to the antenna reduces these losses, as shorter guided cables are used. However, in cases where the RRU is too heavy to be mounted near the antenna, it may still be placed at the base of the tower, with appropriate adjustments to mitigate losses.

Figure 4.5: RRU

Baseband Unit (BBU)

The BBU is the brain of the Base Station System (BSS). Its primary functions include: Baseband signal processing, Resource allocation and management and Data routing between RRU and core network.

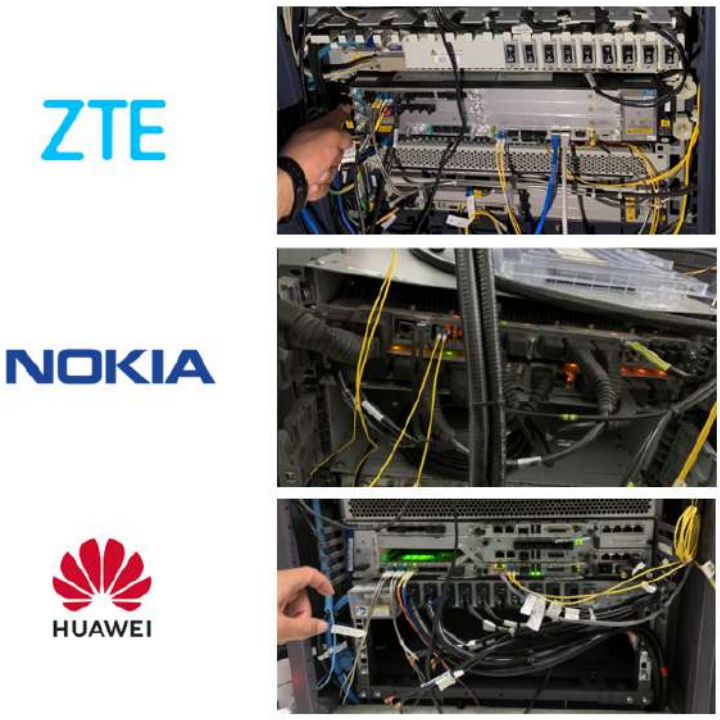

Figure 4.6: BBU : Multi vendor comparison

The BBU and RRU work in conjunction to ensure efficient signal processing and transmission, forming the backbone of modern wireless networks.

4.1.3 The Complete BSS Structure

The Base Station System (BSS) consists of the Baseband Unit (BBU) for signal processing, the Remote Radio Unit (RRU) for amplification and modulation, and the Antenna for wireless communication.

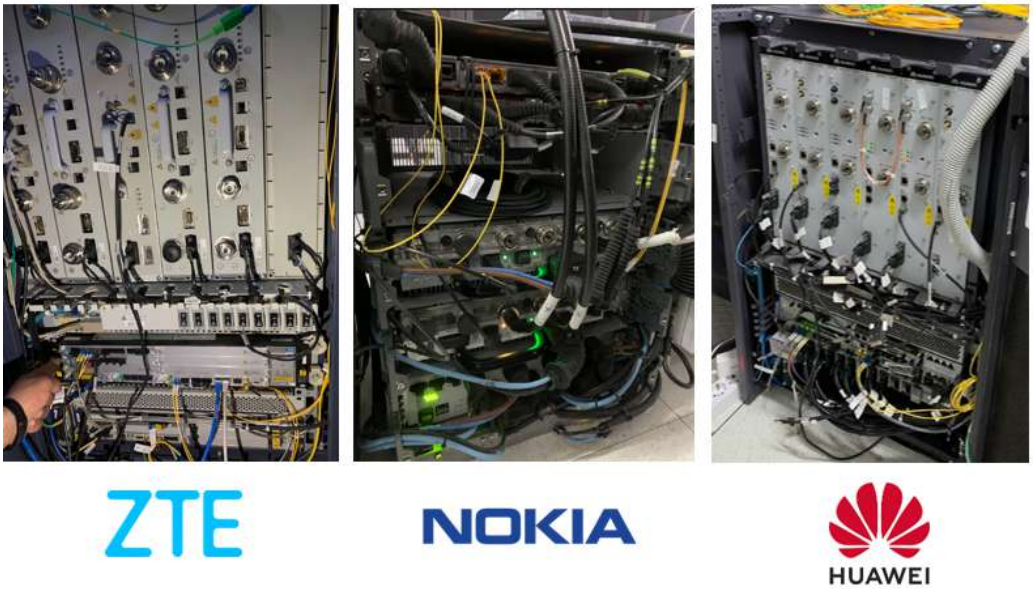

Figure 4.7: BSS: Multi-vendor comparison.

Comparison Between BSS Vendors

The table below provides a comparison of ZTE, Nokia, and Huawei based on their visible BSS designs and observed performance characteristics:

Vendor	BBU Design	RRU Design	Observed Characteristics
ZTE	Compact and organized layout with minimal cabling.	Basic, straightforward connections.	Efficient design, suitable for simpler setups. Maintenance appears to be easy due to its clear structure.
Nokia	Densely packed, with a complex cable arrangement.	Well-integrated, supporting multiple connections.	Focus on robustness and high performance. However, the dense cabling may make troubleshooting more challenging.
Huawei	Balanced modular design with well-organized cabling.	Efficient and flexible RRU connections.	Combines scalability with neatness, allowing for easier maintenance and future upgrades.

Table 4.1: Comparison of BSS Vendors: ZTE, Nokia, and Huawei.

Key Observations

- **ZTE**: The compact and clean layout, with minimal cabling, indicates that ZTE's equipment prioritizes simplicity and cost-effectiveness. It appears most suitable for environments where ease of deployment and maintenance are key.
- **Nokia**: The dense cabling and packed design suggest a focus on performance and redundancy. However, the complex arrangement may require more effort for maintenance and fault analysis.
- **Huawei**: The balanced design, with organized cabling, reflects a combination of scalability and efficiency. This makes Huawei's solution suitable for both current and future network demands.

4.1.4 Different Antennas from the Lab

Figure 4.8: Quad-Antenna Configuration

Figure 4.9: Dipole Antenna

Figure 4.10: Surecall Wide Band Omni-Directional Ceiling Mount Dome Antenna

Figure 4.11: Antenna Tree Configuration: ZTE ZXSDR R8108 T2023 Pico RRU

4.2 MSC (Mobile Switching Center) Visit

During our visit to the Mobile Switching Center (MSC), we observed key components crucial for network management,Below are some pictures capturing these elements.

Figure 4.12: Antennas MSC Station

Figure 4.13: Triple Band Combiners

Figure 4.14: Wirings to the BBU

Figure 4.15: Switches

Chapter 5

Conclusion

This internship was a valuable opportunity to explore the company's structure and its technical operations up close. By working across different departments, we gained a clearer understanding of how each team contributes to maintaining a reliable and efficient mobile network in Algeria.

We delved into the key aspects of network infrastructure, including frequency bands, bandwidth allocations, and the evolution of mobile technology across generations. We also observed how optimization teams detect and resolve network issues, how radio coverage is analyzed, and how transmission and IP core systems function.

Visiting the laboratory and the Mobile Switching Center (MSC) brought a hands-on dimension to our learning. Seeing real-world applications of theoretical concepts, from traditional hardware setups to modern software-driven networks, was particularly insightful.

Overall, this experience deepened our understanding of mobile networks, bridging theory with practice. The knowledge and skills gained will undoubtedly be useful in our future work in the telecom field.

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